Mohammad Javadi Balakan

Experimental Physicist · Quantum Devices · Nanofabrication

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SUMMARY

Experimental physicist with more than a decade of hands-on work taking quantum and semiconductor devices from design and cleanroom fabrication to low-noise cryogenic measurement. Recent work on topological superconductivity in Fe(Te,Se) nanoscale ring devices for fault-tolerant quantum computing. Earlier, CMOS-compatible 2D-material and hybrid organic–silicon optoelectronic devices. Now combining this experience with AI for science and technology.

CORE SKILLS

Device fabrication and structural characterization: cleanroom nanofabrication, process development, electron-beam lithography (EBL), focused ion beam (FIB), scanning electron microscopy (SEM), atomic force microscopy (AFM), reactive ion etching (RIE), Raman and optical spectroscopy, thin film deposition (PVD, e-beam evaporation, CVD, PECVD), 2D materials, wide bandgap oxides, organic semiconductors, topological insulators, and topological superconductors.

Electrical characterization: DC/AC semiconductor parameter analysis (SMU, LCR, etc.), optoelectronic characterization tests (photovoltaic cells, optical detectors/sensors), low-noise/precision measurements (lock-in amplifiers, Atto-sense modules, low-temperature LPFs), cryogenic systems, quantum transport, phase-sensitive magnetic/flux measurements.

Computation, data analysis, and automation: Python, Mathematica, LabVIEW, KLayout (layout/mask design), statistical analysis, instrument automation.

EXPERIENCE

Research Scientist · Stealth-mode startup, Bay Area, CA — 2026 – present

Research Scientist; Postdoctoral Fellow · CNSE, SUNY Albany · Prof. Ji Ung Lee — 2022 – 2026

- Built an end-to-end fabrication process for hBN-encapsulated, nanoscale superconducting ring devices (EBL, FIB, RIE, thin-film deposition).
- Upgraded a closed-cycle cryostat and designed cryogenic low-pass filters; automated low-noise, phase-sensitive measurements in Python.
- Demonstrated half-integer flux quantization, coherent edge transport and intrinsic magnetization of the condensate in Fe(Te,Se); 4 first-author papers and preprints.
- Collaborated with Brookhaven National Laboratory, Stony Brook University and NY Creates; invited talks at LPS and the Gordon Research Conference.

Postdoctoral Associate · Department of Physics, University of Tehran — 2019 – 2021

- First chemical-vapor-deposition growth of borophene, a one-atom-thick metal for 2D interconnects (ACS AMI, 100+ citations).
- Measured carrier transfer across graphene–silicon junctions (10^6× slower than Au–Si) and derived the 25.5% efficiency limit of graphene/Si solar cells.
- Developed position-sensitive detectors and self-powered photodiodes on PEDOT:PSS/silicon (APL Editor's Pick; 10^6 V/W responsivity).

Visiting Researcher · Sapienza Università di Roma · Prof. Giorgio Parisi (2021 Nobel Laureate in Physics) — 2018

Graduate Researcher · University of Tehran; Tarbiat Modares University — 2012 – 2019

- Nanostructured solar cells and photodetectors; Monte Carlo and COMSOL modeling of charge transport and plasma growth.

SELECTED PUBLICATIONS

- **M. J. Balakan**, et al., "Intrinsic magnetization of the superconducting condensate in Fe(Te,Se)," *Physical Review B* (Letter), accepted 2026. **EDITORS' SUGGESTION**
- **M. J. Balakan**, et al., "Half-integer winding of order parameter in an iron-based superconductor," submitted (2026).
- **M. J. Balakan**, et al., "Aharonov–Bohm interference from coherent spin-polarized edge transport in Fe(Te,Se) superconducting rings," submitted (2025), arXiv:2508.05030.
- A. Mazaheri, **M. Javadi**, Y. Abdi, "Chemical vapor deposition of two-dimensional boron sheets by thermal decomposition of diborane," *ACS Applied Materials & Interfaces* 13(7) (2021).
- **M. Javadi**, AA. Noroozi, Y. Abdi, "Kinetics of charge carriers across a graphene–silicon Schottky junction," *Physical Review Applied* 14 (2020).
- **M. Javadi**, M. Gholami, H. Torbatiyan, Y. Abdi, "Hybrid organic/inorganic position-sensitive detectors based on PEDOT:PSS/n-Si," *Applied Physics Letters* 112 (2018). **EDITOR'S PICK**

23 publications and preprints · 437 citations · h-index 13 (Google Scholar). Full list at www.mjbalakan.com. Earlier papers under M. Javadi.

EDUCATION

PhD in Solid-State Physics, University of Tehran, Iran — 2019

MSc in Nanoscience and Nanotechnology, University of Tehran, Iran — 2014

BSc in Physics, Urmia University, Iran — 2012

HONORS

Chamran Postdoctoral Fellowship — 2020

Outstanding PhD Thesis in Physics (nationwide) — 2019

Outstanding PhD Dissertation in Physical Sciences — 2019

INEF Graduate Scholarship — 2016–2018

TALKS AND SERVICE

Invited talk, Gordon Research Conference (Correlated Electron Systems) — 2026

Talk, APS Global Physics Summit — 2026

Invited talk, Laboratory for Physical Sciences, College Park, MD — 2025

Referee for Physical Review Letters, Applied Physics Letters, 2D Materials, Nanotechnology